

Gamification in Education

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September 28, 2026

1 Introduction

Once, we learned for knowledge.

Then came the points.

Then the badges.

Then the leaderboards.

And suddenly, education was no longer about learning.

It was about XP.

Gamification has become increasingly common in education, with points, badges, levels and rewards being used to motivate students and increase engagement. For Ester, however, gamification is not something she is particularly fond of. Having worked in a school and as a private tutor, Ester has seen how deeply gamification can influence the learning experience. Her father is also a teacher and frequently talks about the use of different methods to engage students.

Today, Ester works as a consultant at a company that builds Salesforce CRM solutions. She has always been the kind of person who learns best on her own terms: kind and patient with others, but stubborn when someone tries to tell her how she ought to study. When her employer began pushing her to pass the Salesforce Administrator certification, she took the pressure seriously and put in the hours, but the badges, points and progress bars built into the learning platform never motivated her the way they were meant to. If anything, they made a subject she already respected feel like something being performed for a scoreboard rather than something she was learning for herself. Gamification is therefore no longer limited to games. It has become part of education, professional development and everyday learning. This raises an important question: does gamification actually improve learning, or does it simply make education more engaging? Reviews of the field suggest that gamification does have positive cognitive, emotional and behavioural effects overall, but that the quality of the design and the specific game elements used matter as much as the presence of gamification itself [1].

2 The Battle of the Gamers

Ester is sitting at home, the room lit only by the pale glow of her laptop screen. On the desk beside her, a can of soda has gone flat, forgotten somewhere between the third and fourth round. The notification sound plays again, a short bright chime, the same chime that plays every time someone's name moves. She has learned to tell the difference between the chime that means someone climbed and the one that means someone fell, though the sound itself never changes, only her stomach does. On the screen, a leaderboard refreshes itself in real time. Names rearrange, numbers tick upward in green, small animated stars burst beside whoever sits at the top. Her own name sits near the bottom, grey against all that shine, and no matter how quickly she answers, no matter how many rounds she plays, it barely moves. Through the wall she can hear her brother in the next room, laughing, his keyboard clattering fast and easy, his name climbing without him even seeming to try. Downstairs, her mother calls up to ask how she is doing, and Ester types back that she is doing great, thumbs quick over the keys, the lie smoother than she expected it to be. Her palms are damp against the trackpad. Her jaw aches from smiling at a screen no one else can see. Every few minutes a new round begins, and the timer counts down in the corner, red numbers shrinking second by second, and her hand hovers over the keys, already bracing for the small drop her name will take before the numbers even land.

Somewhere close to midnight the chime rings one more time and Ester does not hear it. Her head has come to rest against her forearm, the cursor blinking uselessly in an empty answer field, and the room around her dissolves the way rooms do in sleep: quietly, without a single seam to mark where waking ends. She opens her eyes onto a stadium she has never been to, tiered seats rising into a sky the exact yellow of a Salesforce onboarding screen, and she knows, the way you sometimes know things in dreams without being told, that she is dreaming, and that this particular dream is not going to let her go until she has earned it.

Astro is waiting for her at the centre of the field. Not the small plush toy from the trade show booths, but life sized, a raccoon zipped inside its own furred hood, round black eyes catching no light at all. Its voice, when it speaks, is warm in the practiced way customer service is warm. Three towers rise out of the grass at its gesture, each one a level Ester will have to clear before she can go home. Argue well, Astro tells her, and she moves on. Fail to convince it, and she stays.

The first tower is labelled Motivation, glowing a faint gold. The second is Control, humming low, its light the exact colour of a progress bar filling. The third Ester recognises without needing to be told, because she has been living inside it all evening: the Leaderboard, tall and mirrored, throwing her own reflection back at her from every angle she turns. Astro gestures toward the first tower and waits, patient in the way only something built to wait can be.

"Whenever you are ready," it says.

Ester does not move yet. She is still counting the towers, still working out

which argument she is going to need first, and for now that is where the dream leaves her: standing on the grass of a stadium built out of onboarding screens, three unclimbed towers ahead of her, and an AI raccoon waiting patiently for her to begin.

References

- [1] Rabah, Cassidy, and Beauchemin, “Gamification in education: Real benefits or edutainment?,” in *European Conference on eLearning (ECEL)*, 2018.
- [2] Gomez, Moron, Ferrandez, Saiz, and Garrido, “Gamification in the education: The benefits of learning through play,” in *INTED2017 Proceedings*, IATED, 2017.
- [3] Chapman and Rich, “Does educational gamification improve students’ motivation? if so, which game elements work best?,” *Journal of Education for Business*, 2018.
- [4] Nistor and Iacob, “The advantages of gamification and game based learning and their benefits in the development of education,” in *International Scientific Conference eLearning and Software for Education (eLSE)*, 2018.
- [5] Romano Daba, Rosmansyah, and Dabarsyah, “Problem based learning using gamification: A systematic literature review,” in *IEEE International Conference on Information and Communication Technology and Systems (ICIMCIS)*, 2019.
- [6] Harviainen, “Critical challenges to gamifying education: A review of central concepts,” in *Game On! Conference*, (Moscow), 2014.
- [7] Biryukov, Brikoshina, Mikhalevich, Sycheva, and Khalimon, “Gamification in education: threats or new opportunities,” in *SHS Web of Conferences*, 2021.
- [8] Mirzaie Feiz Abadi, Khalili Samani, Akhlaghi, Najibi, and Bolourian, “Pros and cons of tomorrow’s learning: A review of literature of gamification in education context,” *Medical Education Bulletin*, 2022.
- [9] Toda, Valle, and Isotani, “The dark side of gamification: An overview of negative effects of gamification in education,” in *Higher Education for All (HEFA) Workshop*, Communications in Computer and Information Science, Springer, 2018.
- [10] Thorsteinsson and Niculescu, “Pedagogical insights into the use of minecraft within educational settings,” *Studies in Informatics and Control*, 2016.

- [11] Slattery, Butler, O’Leary, and Marshall, “Primary school students’ experiences using Minecraft education during a national project based initiative: An Irish study,” *TechTrends*, 2023.
- [12] Baek, Min, and Yun, “Mining educational implications of Minecraft,” *Computers in the Schools*, 2020.
- [13] Kuutti, “Designing gamification,” Master’s thesis, Oulu Business School, 2016.
- [14] Steinbeiss, “Minecraft as a learning and teaching tool: Designing integrated game experiences for formal and informal learning activities,” Master’s thesis, University of Oulu, 2017.
- [15] Caponetto, Earp, and Ott, “Gamification and education: A literature review,” in *8th European Conference on Games Based Learning (ECGBL)*, 2014.
- [16] Christopoulos and Mystakidis, “Gamification in education,” *Encyclopedia*, 2023.
- [17] Dicheva, Dichev, Agre, and Angelova, “Gamification in education: A systematic mapping study,” *Educational Technology & Society*, 2015.
- [18] Huang and Soman, “A practitioner’s guide to gamification of education,” tech. rep., Behavioural Economics in Action, Rotman School of Management, University of Toronto, 2013.
- [19] Ofosu Ampong, “The shift to gamification in education: A review on dominant issues,” *Journal of Educational Technology Systems*, 2020.
- [20] Kim and Werbach, “More than just a game: Ethical issues in gamification,” *Ethics and Information Technology*, 2016.
- [21] Khoshnoodifar, Ashouri, and Taheri, “Effectiveness of gamification in enhancing learning and attitudes: A study of statistics education for health school students,” *Journal of Advances in Medical Education and Professionalism*, 2023.